PUBLIC PORTFOLIO TRANSLATION

Industrial Predictive Model

Validation and Reviewer Response Q&A

Document type: English translation of redacted Greek technical Q&A documentation.

Project context: Professional M&V project - industrial energy baseline modeling and Measurement & Verification case study.

Use: Public portfolio documentation for the Industrial Energy Forecasting Platform repository.

> **Note.** This is an English translation prepared for public portfolio review. The original Greek redacted documentation remains the authoritative case-study evidence. Sensitive operational data and private details are not included.

Industrial Predictive Model - Validation and Reviewer Response Q&A

> English translation prepared for the public portfolio repository.
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> This document is a translated version of the Greek technical response document. It addresses reviewer-style questions about overfitting, validation leakage, reporting-period MAE, uncertainty, multicollinearity, hyperparameters, random seed sensitivity, and robustness of the savings conclusion.

1. Model reliability: overfitting and validation

A. Comment on the high training R^2

Reviewer comment:

> "The fact that the model achieves almost 100% accuracy ($R^2 = 0.994$) during training usually means it has memorized the data…"

This observation mainly applies to linear or heavily over-parameterized models. In the case of tree-based boosting models such as CatBoost, a very high training R^2 is expected when the model correctly learns the structure of the data.

The critical question is not whether the training R^2 is high, but whether:

- the model maintains good generalization on independent data; and
- the model remains stable after the intervention.

This is evaluated through:

- the validation period within the baseline period; and
- the reporting period after the intervention.

The results were:

Period	R^2
Training	0.994
Reporting	0.925

The preservation of high R^2 during the reporting period shows that the model does not lose structural validity after the intervention and has not merely memorized the training data.

If serious overfitting existed, the reporting-period R^2 would be expected to collapse, for example below 0.5. That is not observed.

In addition, the increase in MAE during the reporting period is expected because actual consumption changes due to the interventions. The baseline model continues to predict the counterfactual consumption, not the new reality.

B. Validation-period check and avoidance of data leakage

Following a related comment, the possibility that the validation period was used during model training was explicitly examined.

An overlap check between the training and validation sets showed:

```
Overlap train/validation rows = 0
```

This confirms that no observation from the validation period (28/11/2024 - 05/12/2024) was included in the training set. The evaluation was therefore performed on data the model had not seen during training.

The claim that the validation period is included in the training data does not apply to the present implementation.

The exclusion of the validation window from the training set is explicit in the code:

```
train_mask = (
    (df[DATE_COL] <= TRAIN_END) &
    ~((df[DATE_COL] >= VAL_START) & (df[DATE_COL] <= VAL_END))
)
```

The `~(...)` expression means that dates within the validation window are removed from the training set before model training.

C. Why validation is inside the baseline period and not after the intervention

The selection of a validation window inside the pre-intervention baseline period was intentional and methodologically necessary.

The post-intervention period has limited duration, approximately 1.5 months, and already includes the effect of the interventions.

Using post-intervention data as validation would create a methodological error because it would allow the model to be evaluated or tuned on data that already contains the savings effect. This would create baseline contamination.

Instead, the validation window was selected within the baseline period in order to:

- evaluate model generalization on unseen data;
- preserve the same operational and energy conditions as the pre-intervention period; and
- ensure that the model describes the facility's normal behavior without influence from the interventions.

This approach is fully aligned with the principles of IPMVP Option C.

2. MAE, representativeness, and interpretation of the approximately 11% result

This section explains why the claim that "the increase in MAE shows model failure" is not justified in the context of M&V / IPMVP Option C.

2.1 What the increase in reporting-period MAE means

The model was trained exclusively on pre-intervention baseline data in order to estimate the counterfactual consumption: the expected consumption without interventions, as a function of production variables, operational variables, and calendar effects.

After implementation of the energy-saving measures, actual consumption shifts downward, while the baseline model correctly continues to predict the consumption that would have occurred without the intervention.

In this situation, it is expected that the absolute daily deviation (MAE) increases, precisely because the model should not adapt to the new post-intervention reality. If it did adapt, there would be a risk of baseline contamination.

Therefore, the increase in reporting-period MAE is not by itself evidence that the model is non-representative. It is compatible with the presence of a level shift in energy consumption, which is exactly what M&V attempts to quantify: the difference between baseline and actual consumption.

It should also be noted that the period in which MAE increases coincides with installation of the energy-saving measures, not simply with meter installation. In other words, this is the period in which the physical consumption of the factory was intentionally shifted downward.

In IPMVP Option C, the baseline model is trained on pre-intervention data to estimate the consumption that would have occurred without the project. When the saving measures are applied, actual consumption shifts, while the baseline continues to correctly predict the counterfactual.

Under these conditions, the increase in MAE during the reporting period does not indicate model failure. It is the expected numerical expression of the energy level shift caused by the project.

If MAE did not increase after the intervention, that would indicate baseline contamination and underestimation of the savings.

2.2 Distinguishing savings from model failure

The appropriate test is not MAE alone. MAE is a daily error metric. The evaluation must combine several checks.

A. Total energy balance: total deviation / savings percentage

The reported savings are calculated as:

```
Savings % = (sum(Predicted baseline) - sum(Actual)) / sum(Actual)
```

This is not based on isolated daily deviations. It is based on the total energy balance of the reporting period, which is the core logic of IPMVP Option C for whole-facility analysis.

B. Check that the model does not structurally fail after the intervention: reporting-period R^2

During the reporting period, $R^2 = 0.9252$. This means the model continues to explain the structure and variability of consumption in relation to the operational drivers, such as production and calendar effects.

This is important:

- If the approximately 11.9% difference were due to model collapse or non-representativeness, we would expect a strong loss of structural relationship, significantly lower R^2, unsystematic residuals, and unstable behavior.
- Instead, the model maintains high explanatory ability while the consumption level shifts. This pattern is compatible with real consumption reduction after the intervention.

C. Consistency and unbiased behavior during the baseline period

During baseline training:

- MAE = 544.67 kWh
- total deviation = -0.46%
- $R^2 = 0.9940$

The very small total deviation, close to zero, shows that the model does not have systematic bias in the pre-intervention energy balance.

Therefore, when a systematic positive difference appears in the reporting period, where predicted baseline is higher than actual consumption, it is more consistent to interpret this as a change in the level of consumption rather than as random or structural failure of the baseline.

2.3 Why the claim “MAE increased, therefore the model is not representative” is incomplete

That claim confuses two different concepts.

First, MAE is a measure of daily error in absolute units. After the intervention, the apparent error includes the real effect of the intervention, which is not model error but the target result of the analysis.

Second, the representativeness of a baseline model in M&V is judged by its stability and lack of bias during the pre-intervention period, meaning bias close to zero and good model fit, and by whether the model does not structurally collapse during the reporting period, meaning preservation of strong structural explanation such as high R^2 and reasonable residual behavior.

In this case both conditions hold:

- baseline bias: -0.46%, very small;
- reporting R^2: 0.9252, high.

Therefore, the approximately 11.89% difference is interpreted as a level shift, meaning reduced consumption, not as prediction failure.

2.4 Technical conclusion

The increase in MAE during the reporting period is expected and compatible with the presence of real energy savings, because the baseline model predicts consumption without the intervention.

The fact that baseline bias is practically zero and reporting-period R^2 remains high shows that the model remains structurally representative and does not collapse.

Therefore, the observed positive difference in the total energy balance is attributed to a real change in consumption, not to a collapse of predictive capability.

3. Prediction intervals, uncertainty, and IPMVP Option C

In the context of IPMVP Option C, the savings estimate is based on comparison of adjusted baseline consumption with actual measured consumption at whole-facility level.

IPMVP does not require prediction intervals, confidence intervals, or quantile regression to validate a savings estimate unless these are explicitly required by the M&V plan.

The statistical logic of Option C is not to test whether each daily measurement lies inside an uncertainty interval. Instead, the question is whether total reporting-period consumption, normalized to actual operating conditions through a baseline model, systematically differs from the consumption expected without the project.

Methodological clarification

Prediction intervals or quantile regression are suitable when a model is used for forecasting future actual consumption.

In IPMVP Option C, however, the baseline model is not used to predict real future consumption. It is used to estimate the consumption that would have occurred without the project: the counterfactual.

Therefore, the appropriate statistical criterion is not whether reporting-period measurements fall within a prediction interval, but whether total reporting-period consumption differs systematically from adjusted baseline consumption.

In this project, this is achieved through the CatBoost baseline model, which:

- is trained exclusively on pre-intervention data;
- shows low total deviation and no systematic bias during baseline and validation periods;
- is applied to the real production and calendar conditions of the reporting period to generate the "without project" consumption.

Savings are calculated from the energy balance:

```
Savings = Adjusted Baseline - Actual
```

This follows the logic of IPMVP Option C.

Prediction intervals or quantile regression may provide additional analytical information, but they are not an IPMVP requirement for the validity of the savings estimate.

Reliability is based on baseline stability, lack of bias, and adequate fit. These criteria are satisfied by the present system.

Additional uncertainty analysis

Although the above is fully sufficient under IPMVP and the project contract, an additional bootstrap analysis was implemented for completeness and stronger statistical documentation.

The analysis was applied to the reporting period by resampling the daily pairs of predicted baseline and actual consumption with replacement. This produced a distribution of possible total-savings values.

In each bootstrap iteration, savings were calculated using the same definition applied under IPMVP Option C.

The analysis was performed on a conservative model in which:

- all variables with high multicollinearity were removed, with VIF < 5 for all variables;
- a validation window closer to the reporting period and longer than one month was used, in order to test baseline stability under stricter generalization conditions:

```
VAL_START = pd.to_datetime("2025-04-28")
VAL_END = pd.to_datetime("2025-05-28")
```

The bootstrap analysis results, at 90% confidence level, were:

- Total savings: 249,233 kWh
- 90% confidence interval in kWh: [166,526, 337,385]
- Estimated savings percentage: 14.70%
- 90% confidence interval in percentage terms: [8.88%, 22.65%]

At the same time, the structural quality of the model during the reporting period remained high:

- R^2 (Reporting) = 0.862, clearly above acceptable thresholds greater than 0.75 for a stable Option C baseline model.

Conclusion

Even under this more conservative statistical approach, involving removal of correlated variables, a validation window closer to the reporting period, and bootstrap uncertainty analysis, the result remains consistent with the original savings estimate.

The model retains structural stability.

Therefore, under both IPMVP Option C and additional statistical uncertainty checks, the observed reduction in consumption is attributed to real savings rather than statistical noise or baseline-model failure.

4. Correlation, VIF, and interpretation of the cause of savings

It is correct that high correlation between some production variables, such as kg, pallets, and operating hours, makes it difficult to strictly isolate the individual effect of each variable.

However, this issue mainly concerns linear or parametric models where coefficients are interpreted causally. It does not affect the main objective of this project.

In M&V / IPMVP Option C, the goal is not to attribute savings to one specific variable. The goal is to estimate the consumption that would have been expected without the project under the same real operating conditions of the reporting period.

In other words, the baseline model observes the actual production and calendar conditions of the reporting period and estimates what consumption would have been if the project had not been implemented.

Savings are calculated from the facility-level energy balance:

```
Adjusted Baseline - Actual
```

Therefore, even if two variables are correlated, the M&V question is not "which variable caused the reduction?" The question is: "Given these operating conditions, what would consumption have been without the project?"

From this perspective, variable correlation does not invalidate the baseline logic, especially for tree-based and boosting models such as CatBoost, where the model is not based on linear coefficient estimation.

To address the practical concern that the reduction might be caused by a change in production regime, the following points are important:

- the main operational drivers, such as production, operating hours, pallets, and calendar variables, were included precisely to normalize consumption against possible regime changes;
- a sensitivity analysis was also performed by removing the most correlated variables, with all VIF values below 5, and the savings result remained consistent.

This confirms that the conclusion does not depend on one specific correlated variable.

In summary, multicollinearity may limit causal interpretation per feature, but it does not invalidate the logic or reliability of the Option C savings estimate, which is based on adjusted baseline comparison at whole-facility level.

If the goal were to causally attribute savings to individual variables, such as the percentage due to load, operating hours, or other indicators, a different methodological design would be required. This could involve experimental design, design of experiments, or causal modeling.

That is a separate project objective.

In IPMVP Option C and in the present contract, the objective is not decomposition of savings by variable. The objective is reliable estimation of total counterfactual facility consumption and comparison with measured consumption.

That is exactly what the present baseline model implements.

5. Hyperparameters, random seed, and model stability

The reviewer question focuses on whether the estimated savings result could be an artefact of specific hyperparameter choices or the algorithm's random seed.

This concern is reasonable in forecasting applications. However, in IPMVP Option C, the critical objective is not optimal prediction in general machine-learning terms. The critical objective is baseline stability, lack of bias, and temporal consistency when the model is applied after the intervention.

5.1 Role of hyperparameters in IPMVP Option C

The hyperparameters used in the CatBoost model were:

- learning_rate = 0.18
- iterations = 185
- depth = 6

These values are within the recommended ranges from the literature and official CatBoost documentation for general regression models.

For gradient-boosted tree models such as CatBoost, the literature and documentation typically recommend learning rates in the range of 0.01 to 0.3, tree depths usually between 4 and 10, and a number of trees from roughly 100 to several hundred. Complexity is mainly controlled through the combination of learning rate and iterations.

The values used here fall within accepted ranges and correspond to a controlled-complexity model suitable for M&V applications.

In gradient boosting, a useful approximation is:

```
learning_rate x iterations ≈ total model complexity
```

The combination of a moderately high learning rate, a limited number of trees, and moderate depth produces a controlled-complexity ensemble.

This avoids both overfitting and excessive sensitivity to random fluctuations in the data.

This configuration is fully compatible with M&V logic, where baseline stability is more important than maximizing in-sample accuracy.

5.2 Why automated optimization such as cross-validation or Optuna is not appropriate here

Hyperparameter optimization through cross-validation or Optuna maximizes statistical accuracy inside the baseline dataset.

In IPMVP applications, this can often produce overfitted models that learn noise in correlated production variables such as kg, pallets, and operating hours.

This can reduce the model’s ability to function as a stable counterfactual after the intervention.

Therefore, selected, documented, and conservative hyperparameters are methodologically preferable for M&V compared with automated Kaggle-style optimization.

5.3 How sensitivity was tested in practice

Instead of a simple random-seed sweep, a stricter stability check was applied. This included:

- removal of correlated variables, with VIF < 5 for all inputs;
- a new validation window closer to the reporting period and longer than one month: 28/04/2025 - 28/05/2025;
- bootstrap resampling of reporting-period pairs of predicted baseline and actual consumption.

This combination tests:

- the effect of multicollinearity;
- temporal generalization of the model;
- uncertainty due to data and model parametrization.

If the result were an artefact of specific hyperparameters or random seed, this would appear as:

- a very wide bootstrap confidence interval; or
- a significant collapse of reporting-period R^2.

Instead, the observed results were:

- R^2 (Reporting) = 0.862, clearly above acceptable Option C thresholds greater than 0.75;
- a consistent estimated savings level even under the conservative model.

As documented above, reliability under IPMVP Option C does not depend on hyperparameter tuning or random seeds alone. It depends on baseline stability, absence of bias, and structural consistency after the intervention.

These were tested through variable decorrelation, stricter validation, and bootstrap analysis.

Nevertheless, to explicitly address the reviewer's request and exclude the possibility that the result is an artefact of specific hyperparameters, an additional empirical check was performed using two intentionally different model configurations with opposite complexity philosophies.

5.4 Empirical check with different hyperparameters

A. Conservative model with low complexity

Parameters:

```
iterations = 400
learning_rate = 0.15
depth = 4
random_seed = 42
```

Results:

- Baseline MAE = 872.80 kWh
- Baseline deviation = -0.69%
- Validation period = 28/04/2025 - 28/05/2025

- Validation MAE = 703.20 kWh
- Validation deviation = 0.48%
- Reporting MAE = 5,926.60 kWh
- Reporting deviation / savings = 10.90%

The model remains unbiased during validation and produces a significant savings estimate.

B. More flexible model with increased complexity

Parameters:

```
iterations = 400
learning_rate = 0.10
depth = 10
random_seed = 42
```

Results:

- Baseline MAE = 48.8 kWh
- Baseline deviation = -0.11%
- Validation MAE = 1,005.4 kWh
- Validation deviation = 7.89%
- Reporting deviation / savings = 12.36%

The more aggressive model overfits the baseline period, as shown by the extremely low MAE, and shows weaker validation generalization.

Nevertheless, it still indicates strong positive savings during the reporting period.

5.5 Conclusion

Even under radically different parameterizations, conservative versus flexible, the estimated savings level remains consistently in the range of 11% to 12%.

This demonstrates that the result is not an artefact of the hyperparameters or random seed. It reflects a real change in consumption after the intervention.

The hyperparameters are not a source of result instability. They were selected within documented ranges, with controlled complexity and methodological consistency with IPMVP Option C.

The robustness of the result was confirmed through stricter structural checks: variable decorrelation, stricter validation, and bootstrap analysis. These are substantially stronger than simply changing random seed or tree depth.

Therefore, the estimated savings are not an artefact of hyperparameters. They reflect a real change in consumption after the intervention.

End of English translation.